IMPLEMENTATION OF DIGITAL APPROACHES IN BUSINESS PROCESSES OF OIL REFINERIES

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ABSTRACT

Rapidly changing market conditions dictate new requirements for the organization of production enterprises. This is particularly true for high-tech industries such as oil refining.

With a view of maintenance of competitiveness of work of the oil refining enterprises, it is expedient to aspire to increase of level of automation of technological and business processes that it is possible to realize by introduction in architecture of information systems of refinery IS of classes ERP, MES and ACSPP. To ensure effective integration of these IS data with the current information environment of the enterprise, the right solution will be to re-engineer some basic business processes. These measures will improve the quality and depth of processing of raw materials through the transition to automated systems that allow real-time adjustment of equipment operation modes depending on the established planned needs for commercial products and technological limitations.

At carrying out of these actions it is necessary to consider the branch features inherent in difficult processes of oil refining manufacture which are also described in given article.

CCS CONCEPTS

• Applied computing → Enterprise computing → Enterprise architectures • Applied computing → Enterprise computing → Business process management

DTMIS '20, November 18–19, 2020, Saint Petersburg, Russian Federation

ACM ISBN 978-1-4503-8890-0/20/11…$15.00
https://doi.org/10.1145/3446434.3446477

KEYWORDS

Enterprise architecture, Business Process reengineering, Refining, Refineries, Refining business processes, Architectural approach

ACM Reference format:

Roman Marchenko, Alissa Dubgorn, Anastasia Babyr and Berry Gerrits. 2020. IMPLEMENTATION OF DIGITAL APPROACHES IN BUSINESS PROCESSES OF OIL REFINERIES. *In SPBPU DTMIS '20: Proceedings of Peter the Great St. Petersburg Polytechnic University International Scientific Conference "Digital Transformation on Manufacturing, Infrastructure and Service", November 18–19, 2020, Saint – Petersburg, Russia. ACM, New York, NY, USA, 5 pages.* https://doi.org/10.1145/3446434.3446477

1 Introduction

Speaking of one of the most important features of refining business processes, it is necessary to focus on a continuous cycle of refinery production, in which even one day of downtime can lead to significant monetary losses. That is why the production and other processes management system should be designed in such a way as to minimize (exclude) possible downtime. Process management should provide effective information and technological support of the main production and auxiliary processes of the oil refinery.

High volatility of quality and volumes of raw materials in the conditions of existing technological limitations of production equipment causes additional difficulties, which can be reduced or eliminated when building an optimal control system of technological and business processes. The main difficulties in the area of the refinery's material flow control unit are related to the limited possibilities of semi-product quality control on the flow, as usually the quality control is provided by taking from 2 to 3 samples per day. Besides, the majority of refineries have fixed assets with a considerable service life for which significant capital expenditures are required for application of modern technologies, equipment, catalysts, automatic and advanced control systems, which does not allow to quickly and efficiently re-engineer business processes, which requires complex automation of processes, providing the possibility of integration of different class IS into the enterprise IS architecture [1].

It is these difficulties that lead to the fact that decisions related to technological processes and production planning processes at refineries are made in an expert manner, in conditions of insufficient information and transparency of processes, which often leads to the adoption of decisions that are not optimal from the point of view of the target strategy of the enterprise [2].

The expediency of re-engineering the business processes of oil refineries built in the times of the USSR, is due to a greater extent to the historical political and economic situation in the country, when the main monetary, labor and intellectual resources were directed to the oil industry, and oil and fuel oil, which are the further raw materials of deepening processes, were sent for sale abroad. At the same time, fuel oil was consumed in large volumes by energy enterprises and the fleet. That is why the demand for refining capacities, including enriching and deepening processes, was not so high, but at the moment the enterprises are trying to realize the full cycle of oil products reproduction in Russia, excluding the production of fuel oil, which sets us the task of ensuring quality and modern business processes.

The task of this article is to study the industry-specific features of business processes of the refinery, as well as the rationale for the integration of ERP, MES and SCADA systems into the architecture of information systems of refineries in order to increase the level of automation of technological and business processes at the enterprise [3].

2 Materials and methods

Modern refineries have refining capacities, which may include up to 50 different units, each of which is characterized by several operating modes affecting the quality characteristics of the output product. In the conditions of a large number of limitations (from technological limitations of the units themselves, the limitations of planning and production management, regulating the output of commercial products), an important task is the system of monitoring and control over changes in the operating modes of equipment and oil refining units in real time. In order to ensure prompt change of equipment operation modes and control of technological processes of oil refining it is expedient not only to develop a system of key performance indicators (KPI) of production, but also to develop and implement the architecture of information systems of different classes (SCADA, MES, ERP) [4].

All received technological data (quality, equipment operation modes) should be analyzed together with taking into account economic parameters (demand, price, commodity nomenclature), and it is necessary to take into account all types of costs that the company bears when ensuring the operation of production processes (operating costs) and which are taken into account when calculating the cost of final commodity products of the refinery.

The main problem is to ensure the possibility of rapid calculation of the optimal modes of equipment, which provides the solution to the problem of maximizing marginal profit and reducing operating costs with given initial characteristics of the quality and volume of raw materials at the input and rapid response to the physical changes in operating modes [5]. The most effective solution here can be reengineering of business processes of processing, planning and sales in order to implement information systems that ensure the change of equipment operation modes on the basis of calculations of the planning department with regard to restrictions and in real time.

3 Results

As a rule, the nomenclature, quality characteristics and quantity of oil products produced at refineries are regulated by production plans, which are restrictions for a rather complicated task of determining the optimal operating modes of technological facilities and equipment.

The following are some examples of the main industry-specific features of the business processes of oil refining companies, which cause difficulties in the management of refinery processes:

- Continuous production cycle;
- High marketing risks (high volatility of prices for raw materials and processed products);
- Complexity of production planning processes;
- Difficulty in selecting technological modes of equipment operation;
- High dependence on quality and quantity of input raw materials (as a consequence, the need to combine IS with the companies-suppliers of raw materials);
- High level of development of the transport and logistics complex if it is necessary to plan shipments by rail and pipeline in advance;
- High level of dependence on energy resources;
- Peculiarities of the products of oil feedstock processing, etc;
- Necessity to carry out scheduled preventive maintenance works on technological installations.

For effective management of all business processes of the refinery it is necessary to analyze a large amount of data organized into a single dynamic system [6]. Dynamic system implies constant change of process parameters, analytics of which assumes use of essentially new approach to the organization of refinery activity, re-engineering of business processes, introduction of electronic document circulation and development of information systems, allowing to automate production processes and to introduce methods of predictive analytics in order to achieve optimal modes of operation of technological objects of the enterprise.

At the given approach the oil refining enterprise should be considered as the complex system consisting of various functional elements which co-operate among themselves.

Modelling of business processes of oil refinery is expedient to make by means of architectural methods [7].

The architectural method assumes use of concrete principles and techniques by means of which high quality of work of business processes at the enterprise, and also effective integration of existing and new information systems in information structure of the company is provided. In addition, the architectural approach to modeling and re-engineering of business processes allows to achieve the set strategic and operational goals taking into account the use of the information environment and information systems of the enterprise [8].

Often at the industrial enterprises it is not possible to make modernization of current information systems, more and more often the expedient decision is realization of reengineering of business processes, integration of essentially new IS, appendices, services that automatically puts a problem of working out of architectural decisions in the field of management of business processes.

The concept of enterprise architecture is a comprehensive approach to presenting business as a single system of heterogeneous elements - strategic objectives, functional structure, business processes, personnel, information systems and technologies, data, infrastructure [9].

The architectural approach helps to introduce information products and information systems with maximum efficiency and efficiency, as it allows to build exactly that level of communication between business and IT, at which it is possible to implement activities on modeling, implementation and integration of balanced architectural solutions [10].

Using the approach described above, for integration into the existing environment of new information systems at the enterprise, the authors have developed a corresponding functional scheme of typical business processes of the upper level, which are reflected in the structure of the refinery [11]. As a classification of groups of business processes the authors use the classification according to Michael Porter, which is based on the understanding of their role in creating a personal contribution to the company's values. According to this classification, business processes are divided into 3 groups:

- Supporting processes are processes that are designed to serve the main business processes (repair services, logistics, etc.);
- The basic processes - the processes directly connected with a commodity product (logistics, processing, etc.);
- Management processes - processes aimed at strategic activities of the organization, control and monitoring activities, etc.

Figure 1 shows the functional diagram of top-level business processes of a typical refinery, developed by the authors using an architectural approach.

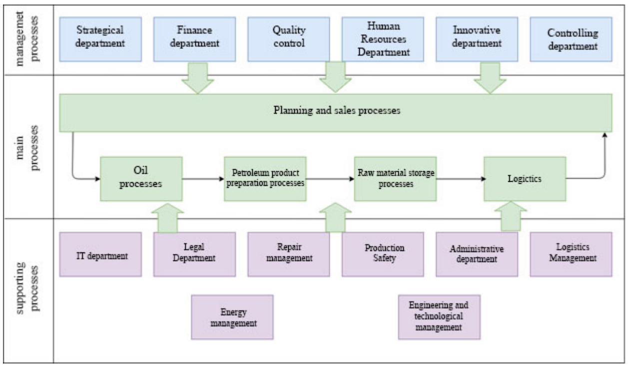

Figure 1: Top-level business process architecture of a typical refinery

With a view of research of branch features of reengineering of business processes of the oil refining enterprise by authors it is offered to consider interaction of the basic processes of refinery - Oil refining, reception of components, processes of planning of manufacture of production, purchase of raw materials and sale of commodity production. All business processes of the refinery are closely connected with each other, and the production processes to which oil refining relates directly depend on the planning and sales processes. Information interaction between the basic business processes is provided by means of IS of class ERP which are not calculated on fast iterative calculation of a considerable quantity of the data arriving in a mode of real time from the equipment and other IS. For performance of functions of optimization of manufacture IS of class MES are applied [12]. Reengineering of business processes of the enterprise is expedient to spend both at introduction of IS in architecture of the enterprise, and at maintenance of correct interaction between systems of a different class. Thus, it is more effective from the point of view of logic of processes to use uniform information model at which the general information space at which is formed, allowing to simplify speed and quality of an information exchange between IS is formed. Also work of responsible employees in different systems is simplified.

Authors offer to consider realization of reengineering of business processes on an example of introduction of systems of class MES and ERP in processes of planning and manufacture.

Planning processes are directly related to the production processes, because it is the planning that determines the target restrictions both on the input volumes and the quality of the incoming raw materials for processing, and on the volume, quality and assortment of final commodity products [10]. It is based on the target restrictions that it is necessary to control the movement of material flows of raw materials and semi-products through all the technological installations included in the process of oil refining, as well as to regulate the modes of operation of equipment (with changes in which it is possible to obtain different characteristics of the required marketable products).

In general, the scheme of movement of material flow of oil raw materials and semi-products at refineries can be presented in the form of the scheme shown in Figure 2.

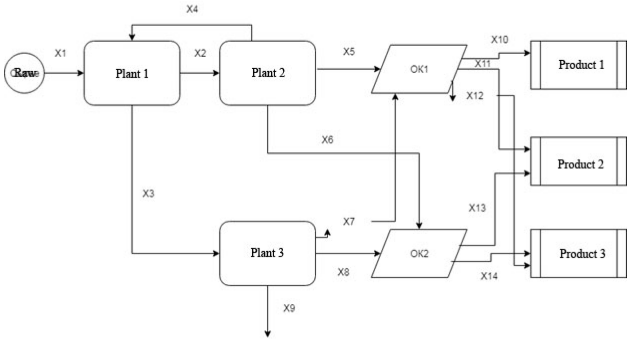

Figure 2: Material flow (raw material) flow diagram

The main purpose of the refinery's activity is to find the optimal solution in the conditions of dynamically changing parameters of external and internal environment.

The target function of this type of tasks is to maximize the total marginal profit from sales of petroleum products. Restrictions of function in this case will be restrictions of technological installations of processing on quantity and quality of processed raw materials and received products (i.e. on an input and an output), requirements to characteristics of final products: GOST, HUNDRED, TU, and also restrictions on preservation of weight of a material stream of a kind:

$$\begin{aligned} &X1 + X4 = X2 + X3 \\ &X2 = X4 + X5 + X6 \\ &X3 = X7 = X8 = X9 \text{ and etc.,} \end{aligned} \tag{1}$$

Where:

x1 - raw materials (oil)

x2-x14 - semi-products (processed raw materials) and components for the production of marketable products (gasoline, diesel fuel, fuel oil)

OK1,2 - compounding objects (tanks) that accumulate and separate components, which are fed into the input of processed components from which commercial products are being prepared (T1,2,3).

Raw materials supplied to the inlet of the primary processing plant are accepted as a constant actual value. Technological limitations on quality and quantity at all plants are also unchanged. Equipment operation modes may change at process plants in case of changes in requirements to final products. At the moment in a number of oil refining companies the simplified scheme of calculation of plans and choice of operation modes of technological objects and mixing of received semi-finished products is carried out, which cannot guarantee the optimum technology and reception of the planned results on quantity and quality of oil refining products [13].

Since the raw materials supplied to the plants may have different characteristics (in a certain range), depending on the amount of output and production task, and the requirements for the output of products may also vary, we are dealing with certain limitations on both sides and must solve the problem of linear programming with a target function of maximizing marginal profit. The solution proposed by the authors is the introduction of information systems capable of collecting data from different functional units (planning, marketing and sales) to determine the dynamic limitations on volume and quality, then iterative calculation of all functions at each stage:

$$P^{`} = k_1X_1 + k_2X_2 + k_nX_n \rightarrow max, \tag{2}$$

Where:

$P^{`}$ – Target function of margin profit maximization

k_n – price of marketable products n

X_n – volume of marketable products n

In this case, at each stage of semi-product processing and components production (before inputs to Devices 1-n and OK 1-n) it is necessary to perform iterative calculation taking into account technological limitations (values are constant and do not change), and at the output of calculations data on the optimal mode of operation of the plant at each stage should be given.

4 Discussion

The analytical calculation model should take into account the probabilistic distribution of the functional dependence between the change in the operating mode of each unit (change in the temperature mode of equipment) and the change in the characteristics of semi-products at the output. According to the authors of the article, to determine the dependencies it is reasonable to use the accumulated retrospective data on the characteristics of input raw materials and output semi-finished products at a given temperature. If the number of available data on changes in the existing parameters of processes and plants is more than a hundred rows (in the matrix, where the number of variable states of the system acts as rows, and as columns - the parameters of the equipment, as well as the qualitative characteristics of products and raw materials), in this case, it is acceptable to use methods of machine learning, for example, regression models boston house dataset or titanic dataset [14]. It will help to reveal necessary dependences of honey by change of equipment operation modes and characteristics of semi-products.

Taking into account the large number of iterations that will be required to calculate all the functions at each stage, the correct solution will be to calculate the required number of equations in advance and create a database with ready-made solutions functions for each predetermined range of values of the characteristics of raw materials, which will be addressed by the MES system to quickly adjust the operating modes of processing equipment [15].

5 Conclusions

In the article the authors analyzed the oil refining branch from the point of view of the architectural approach, all basic, auxiliary (providing) and operating processes of the top level have been allocated, the mechanism of interaction of the basic processes of processing, planning and sale with a view of realization and introduction of the information systems providing automation of productions is offered.

In our opinion, processes of the organization of planning and sale should define restrictions which are shown to industrial processes regarding the nomenclature of commodity production, its quantity and quality depending on results of the marketing analysis of the market. In turn, SCADA class IS should receive processed data from MES systems to set the optimal operating modes of all processing devices at each stage [16]. It becomes possible only at re-engineering of researched business processes and definition of functionality of implemented IS [17].

ACKNOWLEDGMENTS

The reported study was funded by RSCF according to the research project № 19-18-00452.

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